Do Faithful Explanations Go with Accurate Forecasts? An Audit of Plan-Lift Explanations in Retail Forecasting

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Abstract (draft; the audit results are running). Retail forecasting models that see the price and promotion plan can explain each forecast as a baseline, what would sell with nothing planned, times the lift each part of the plan adds. Merchants read that split as fact ("the promotion added 40 units") and plan on it. We audit these explanations for a world model and a direct lift model on M5, Dominick's and Breakfast at the Frat, every saved model version, with eight checks: properties that should hold by construction, sensitivity to randomised weights, stability across seeds, a price-direction check, agreement with what sold, agreement with LightGBM's on/off contributions and TreeSHAP, a plan read-back from the model's state, and agreement with a measured cause. For the cause we find 54 spells in six Dominick's categories in which one store's regular prices sat 6% to 10% off its zone peers for 12 weeks or more, most of them in 1992 and 1993, when the chain ran pricing experiments. Their measured effect on category units, by difference in differences against same-zone stores, gives a pooled response of -0.35 to a price change, close to the -0.4 the chain's own experiments reported. We then ask whether model versions whose explanations pass more checks also forecast better.

Keywords: retail forecasting; explanation; faithfulness; promotions; price response; world models.

1 Introduction

A retail forecast is used for more than its number. When a merchant plans a promotion, the question is not only how much will sell but how much of it the promotion will add, and whether a deeper price cut would add more. Forecasting models that see the plan can answer that directly: they forecast the item with the plan as made, and again with nothing planned, and the difference is the lift the plan adds. Our world model and lift model (paper 1) are built this way: each forecast is a baseline under the nothing-planned plan times a lift, and the lift is exactly zero when nothing is planned.

Such an explanation is useful only if it can be trusted, and being built into the model does not make it right. The explanation literature separates two properties [1]: an explanation is *faithful* if it reflects what the model does, and *correct* if it reflects what happens in the world. In molecular property prediction, MolSanity found the two come apart: attributions faithful to the model were often anti-aligned with the substructure that determines the property [2]. Retail has a property chemistry lacks: some causes can be measured. Dominick's Finer Foods moved store prices on purpose in the early 1990s [3], so the response of sales to a price change can be estimated from the data and the model's stated response compared with it.

This paper audits the plan-lift explanations of every saved world-model and lift-model version from paper 1 with eight checks (Figure 1) and asks whether versions whose explanations pass more checks also forecast better.

2 Related work

Explanations and their evaluation. Feature attributions such as LIME [4] and SHAP [5], with its exact form for trees [6], are the most used way to explain a model's output [7]. Several studies show that they can fail quietly: saliency maps that barely change when a network's weights are randomised [8], attributions that move with input shifts the model ignores [9], unstable explanations for near-identical inputs [10], and attribution rankings that do not survive removing the features they rank highest [11]. Jacovi and Goldberg [1] set out faithfulness as a separate property from plausibility; Rudin [12] argues for models that are interpretable by construction instead of explained after the fact. Our models are of the second kind for the plan, which lets the audit ask whether built-in explanations behave better than post-hoc ones. For time series, explanation methods and their benchmarks are reviewed by Rojat et al. [13]; Ismail et al. [14] and Tonekaboni et al. [15] show that saliency methods designed for images often fail on temporal data.

Price response as a measured cause. Hoch, Drèze and Purk [3] report Dominick's store-level pricing experiments: in each of 19 categories, stores were assigned to higher, lower or unchanged regular prices for 16 weeks, and category units moved by about 3% for a 10% price

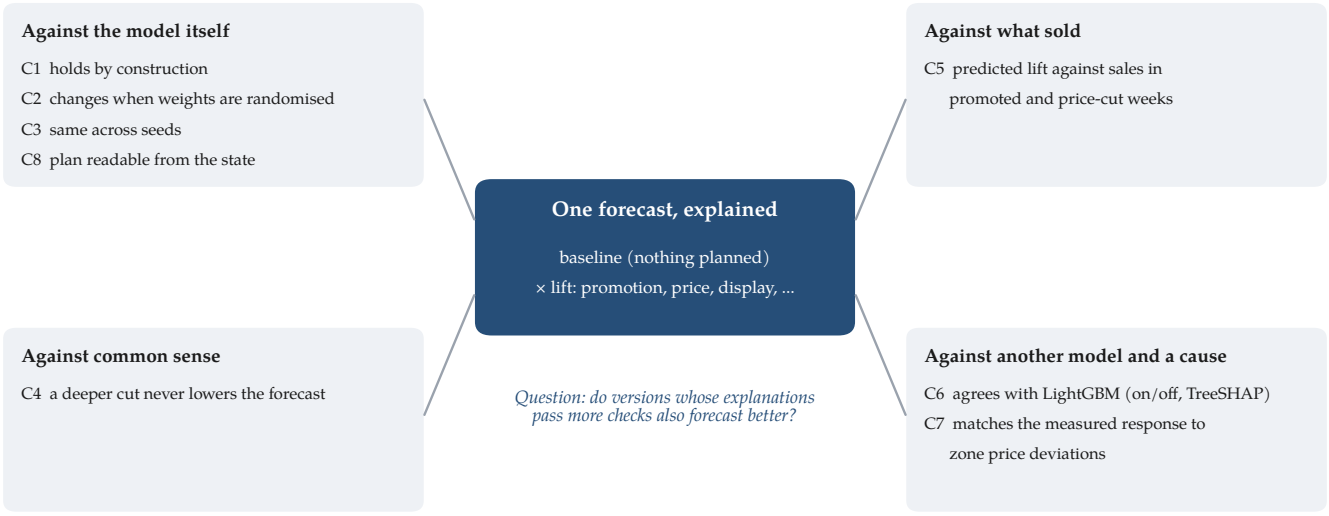


Figure 1: Each forecast's explanation is checked against the model itself, common sense, what sold, another model and a measured cause. The eight checks are fixed in `docs/PAPER2_PLAN.md` before any result; every model version and every check is reported.

change, an elasticity near -0.4. Montgomery [16] and Hoch et al. [17] model store-level price response on the same data. The public files do not record which stores or weeks were in the experiments; Abrams, Gui and Hortaçsu [18] reconstructed them with the original documentation and found that some published price changes did not replicate. We therefore use the price deviations visible in the public data as a quasi-experiment and estimate their effect by difference in differences [19], with standard errors clustered by store [20, 21].

3 What is explained, and the checks

For each item, store and forecast period the audit records the forecast with the plan as made; the forecast with nothing planned (each item at its regular price, its highest over the last 12 periods, and every lever off); and each lever's contribution, the forecast minus the forecast with that lever off and the rest of the plan as made. Contributions need not add up to the total lift; the shortfall is the interaction between levers, reported as such. For LightGBM the same on/off contributions are computed, together with TreeSHAP values on its lever inputs [6].

C1 By construction. Zero lift with nothing planned; a baseline that does not move when the plan changes; baseline plus lift equal to the forecast. Largest violation relative to the largest forecast.

C2 Randomisation. The lift readout, then the baseline readout, the transition and the encoder are re-initialised in turn [8]; rank agreement between the original and randomised lifts should fall towards zero.

C3 Seed stability. Rank agreement of item-week lifts between seeds of the same configuration, set against the agreement of their forecasts.

C4 Price direction. The share of item-weeks where a deeper price cut gives a lower forecast than a shallower one.

C5 Against what sold. On promoted and price-cut item-weeks, the forecast against sales, and the rank agreement of the predicted lift with sales over the model's own baseline. Merchants promote in weeks they expect to sell well, so this measures agreement, not cause.

C6 Against LightGBM. Rank agreement of each lever's contribution with LightGBM's, on/off and TreeSHAP, on the same window.

C7 Against a measured cause. Described in §4.

C8 Plan read-back. From paper 1: the plan read back from the world model's rolled state.

4 Ground truth: zone price deviations on Dominick's

Dominick's set regular prices by zone: within a zone every store was meant to carry the same regular price [22]. For each store and item we take the log of the week's regular price (no deal code, not a temporary cut) over that item's usual regular price in that store, and subtract the median of the same across the zone's stores that week, so moves made zone-wide cancel. An episode is at least 12 consecutive weeks in which this is at least 3% in the same direction for most of the store's items, with

a median of at least 5%; controls are same-zone stores within 1% of their zone over those weeks. The detector was written, and its episodes listed, before any model was scored against them. A first version that measured stores against the zone's most common price found one episode in analgesics: in the experiments a third of a zone's stores moved up and a third down, so the most common price was often a moved one. The change is recorded, with its date, in the plan.

The detector finds 54 episodes in the six categories, 42 with at least three control stores (Figure 2). Most lie between Dominick's weeks 190 and 240, mid 1992 to mid 1993, the period of the chain's experiments; the store prices sit 6% to 10% off their zone peers, in line with the 6% to 24% changes the experiments made. Over the first 8 weeks of each scored episode, category units move against price in 64% of episodes, and 55% have an interval that excludes zero. The pooled slope of the measured response on the price change is -0.35, and the median episode elasticity -0.49, close to the -0.4 Hoch, Drèze and Purk report. Single episodes are noisy, and some move with price, as Abrams, Gui and Hortaçsu found when they replicated the experiments.

For each scored episode the world model and LightGBM are fitted on the data up to the week before it and forecast its first 8 weeks twice: with the store's prices as they were, and with them moved back to its zone peers. The log ratio of the two forecasts of category units is the model's stated response, compared with the measured one.

5 Model versions

Every saved world-model and lift-model checkpoint from paper 1 is audited: the development and final windows of Frat, the development versions on Dominick's, and the M5 practice, confirmation and test windows. The Dominick's final world models, an average of three fits, were not saved in paper 1; they are refitted with settings unchanged for the audit, three seeds each, while paper 1 keeps the scores of its original runs. Each version's accuracy is read from its own run record. LightGBM is refitted on the Dominick's and Frat final windows and the M5 test month for C6.

6 Results

The audit of the model versions (C1 to C6, C8) and the models' stated responses for C7 are running on Colab at the time of this draft; this section is written when they finish.

7 Reproducibility and licence

The checks are in `src/rwm/audit/`, the runs in `scripts/audit_runs.py` and `scripts/paper2_colab.sh`, the plan and its dated amendments in `docs/PAPER2_PLAN.md`, and the episode list in `results/paper2/zone_episodes.csv`. Code under the PolyForm Noncommercial License 1.0.0; text and figures under CC BY-NC 4.0.

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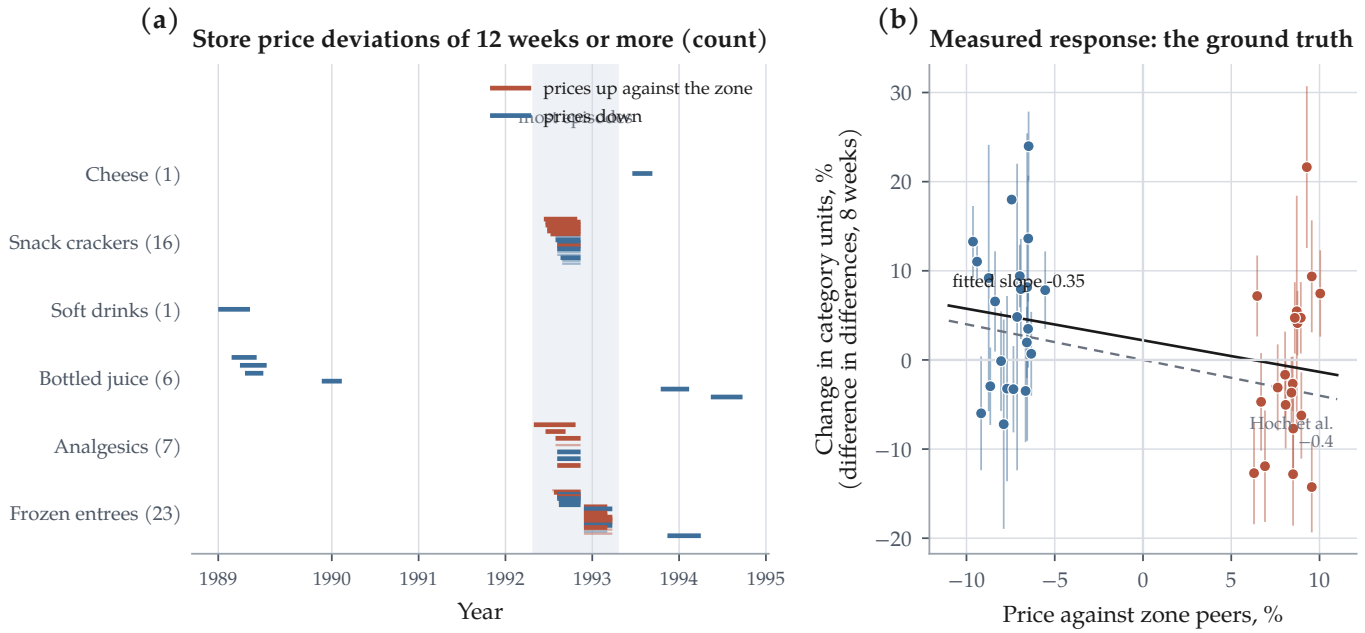


Figure 2: The price deviations cluster in 1992 and 1993, when the chain ran its pricing experiments, and their measured effect on sales matches the experiments' reported response. (**a**) Every episode found in the six categories (count in brackets); faint: fewer than three control stores, listed but not scored. (**b**) For the 42 scored episodes, the store's price against its zone peers and the measured change in its category units over the episode's first 8 weeks (difference in differences against same-zone stores, store and week fixed effects, 95% interval clustered by store). Solid line: fitted slope; dashed: the -0.4 elasticity reported for the experiments [3].

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